Week 0 Final Deliverable – Digital Triage Algorithm

This digital triage system processes patient vital signs and categorises patients into risk levels based on clinically relevant limits. Key parameters including Glasgow Coma Scale (GCS), blood pressure, pulse, temperature, respiratory rate, and blood glucose are all assessed independently. The final risk level is then determined by the most critical finding across all evaluated parameters.

Pseudocode:

```
START

INPUT patient data:
   Age, GCS, SBP, DBP, Pulse, Temperature, Respiratory Rate, Blood Glucose

SET risk_level = "Low"

# Step 1: Check for high-risk/critical conditions
IF GCS ≤ 8 THEN
   risk_level = "High"

IF SBP < 90 THEN
   risk_level = "High"

IF Pulse > 130 OR Pulse < 40 THEN
   risk_level = "High"

IF Temperature > 39 OR Temperature < 35 THEN
   risk_level = "High"

IF Respiratory Rate > 30 OR Respiratory Rate < 8 THEN
   risk_level = "High"

IF Blood Glucose < 70 OR Blood Glucose > 300 THEN
   risk_level = "High"

# Step 2: Check for medium-risk conditions only if not already high
IF risk_level ≠ "High" THEN

   IF GCS BETWEEN 9 AND 13 THEN
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        risk_level = "Medium"

    IF SBP BETWEEN 90 AND 100 THEN
        risk_level = "Medium"

    IF Pulse BETWEEN 100 AND 130 OR Pulse BETWEEN 40 AND 50 THEN
        risk_level = "Medium"

    IF Temperature BETWEEN 38 AND 39 OR Temperature BETWEEN 35 AND 36 THEN
        risk_level = "Medium"

    IF Blood Glucose BETWEEN 70 AND 90 OR Blood Glucose BETWEEN 200 AND 300
THEN
        risk_level = "Medium"

# Step 3: Output final risk level
OUTPUT risk_level

END
```